

Experiment 'p_6_2_a_swap_first_last_array' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_a_swap_first_last_array

Natural language statement: null

Method signature: p_6_2_a_swap_first_last_array(inputs: array<int>) returns (result: array<int>)

Ensures

- `result.Length == inputs.Length`
- `result[0] == inputs[inputs.Length - 1]`
- `result[inputs.Length - 1] == inputs[0]`
- `forall i :: 1 <= i < inputs.Length - 1 ==> result[i] == inputs[i]`

Requires

- `inputs.Length >= 2`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_2_a_swap_first_last_array(inputs: array<int>) returns (result: array<int>)`

The method should respect the following contract:

requires `inputs.Length >= 2`, ensures `result.Length == inputs.Length`, ensures `result[0] ==`
 ↪ `inputs[inputs.Length - 1]`, ensures `result[inputs.Length - 1] == inputs[0]`,
 ↪ ensures forall `i :: 1 <= i < inputs.Length - 1 ==> result[i] == inputs[i]`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnNgzfpqw0lBayP7gOPrUbMnkrbOP

Model: gpt-5.5-2026-04-23

Created at: 1780661517

Finish reason: stop

Usage: 368 tokens in, and 689 tokens out

```
{
  result := new int[inputs.Length];
  var n := inputs.Length;
  result[0] := inputs[n - 1];
  result[n - 1] := inputs[0];

  var i := 1;
  while i < n - 1
    invariant 1 <= i <= n - 1
    invariant result.Length == n
    invariant result[0] == inputs[n - 1]
    invariant result[n - 1] == inputs[0]
    invariant forall j :: 1 <= j < i ==> result[j] == inputs[j]
  {
    result[i] := inputs[i];
    i := i + 1;
  }
}
```

Final Program

```
method p_6_2_a_swap_first_last_array(inputs: array<int>) returns (result: array<int>)
  requires inputs.Length >= 2
  ensures result.Length == inputs.Length
  ensures result[0] == inputs[inputs.Length - 1]
  ensures result[inputs.Length - 1] == inputs[0]
  ensures forall i :: 1 <= i < inputs.Length - 1 ==> result[i] == inputs[i]
{
  result := new int[inputs.Length];
  var n := inputs.Length;
  result[0] := inputs[n - 1];
  result[n - 1] := inputs[0];

  var i := 1;
  while i < n - 1
    invariant 1 <= i <= n - 1
    invariant result.Length == n
    invariant result[0] == inputs[n - 1]
    invariant result[n - 1] == inputs[0]
    invariant forall j :: 1 <= j < i ==> result[j] == inputs[j]
    {
      result[i] := inputs[i];
      i := i + 1;
    }
  }
}
```

Total Token Usage

Input tokens: 368

Output tokens: 689

Reasoning tokens: 512

Sum of 'total tokens': 1057

Experiment Timings

Overall Experiment started at 1780661517349, ended at 1780661560054, lasting 42705ms (42.71 seconds)

Iteration #1 started at 1780661517353, ended at 1780661560054, lasting 42701ms (42.70 seconds)